AI-Based PPE Detection System

An advanced AI-based Personal Protective Equipment (PPE) detection tool

Introduction

Workplace safety is critical in high-risk industries like construction, manufacturing, and chemical processing, where strict use of Personal Protective Equipment (PPE) is essential. However, current manual PPE monitoring methods are labor-intensive, error-prone, and lack real-time responsiveness. To overcome these limitations, this project proposes an AI-Based PPE Detection System. Using computer vision, the system can automatically detect key PPE items—such as helmets, masks, gloves, and safety vests—in real time. It also generates alerts for non-compliance and produces actionable reports. This smart solution enhances safety, reduces human error, and supports a data-driven, proactive safety culture in modern industrial environments.

Why Was This Product Selected

- MANUAL CHECKS FALL SHORT
- TRADITIONAL PPE MONITORING IS **SLOW, INCONSISTENT**, AND CAN'T KEEP UP IN BUSY INDUSTRIAL SETTINGS—LEADING TO SAFETY RISKS.
- REAL-TIME SAFETY IS A MUST
- MOST PEOPLE WE SURVEYED **(83.9%)** BELIEVE AI CAN HELP. EVEN MORE **(93.5%)** WANT REAL-TIME ALERTS TO CATCH ISSUES INSTANTLY.
- DIFFERENT INDUSTRIES, DIFFERENT NEEDS
- SECTORS LIKE CHEMICAL AND FOOD PROCESSING HAVE SPECIFIC PPE RULES THAT NEED SMART, TAILORED SOLUTIONS.
- A GROWING NEED IN BANGLADESH
- WITH RAPID INDUSTRIAL GROWTH AND STRONG AWARENESS OF PPE (93.5% KNOW THE BASICS), NOW’S THE PERFECT TIME FOR THIS TECH.
- SMART DETECTION, SAFER WORKPLACES
- OUR SYSTEM USES AI TO SPOT MISSING GEAR IN REAL TIME—HELPING PREVENT ACCIDENTS AND MAKING SAFETY ENFORCEMENT EASIER.

Feasibility of The Product

1. MARKET AWARENESS & DEMAND

- 93.5% ARE AWARE OF PPE, BUT 67.7% UNAWARE OF AI-BASED PPE DETECTION.
- 83.9% WANT THIS TECHNOLOGY IN THE WORKPLACE.
- TOP PPE NEEDS: HELMETS, GLOVES, AND COMBINATIONS (E.G., HELMETS + MASKS + GOGGLES).
- 83.9% CONSIDER AI-BASED PPE DETECTION IMPORTANT (45.2% "IMPORTANT", 38.7% "VERY IMPORTANT").

2. MARKET SIZE & GROWTH

- GLOBAL PPE MARKET PROJECTED TO REACH $171.66B BY 2034 (CAGR 7.24%).
- BANGLADESH'S DEMAND RISING DUE TO GROWTH IN CONSTRUCTION, MANUFACTURING, HEALTHCARE.
- TAM: $1.2B – $1.5B IN BANGLADESH.
- SAM: $360M – $450M (HIGH-SAFETY INDUSTRIES).
- SOM: $3.6M – $22.5M (REALISTIC SHORT-TERM MARKET SHARE).

MARKET ANALYSIS

Potential Customers:

- Construction companies – high-risk sites needing constant PPE monitoring
- Manufacturing industries – machinery-heavy environments with safety compliance needs
- Chemical and pharmaceutical plants – require strict and specific PPE enforcement
- Shipbuilding yards – large areas where manual PPE checks are impractical

Market Analysis:

Bangladesh Market:

- TAM: USD 1.2 – 1.5 billion
- SAM: USD 360 – 450 million
- SOM: USD 3.6 – 22.5 million

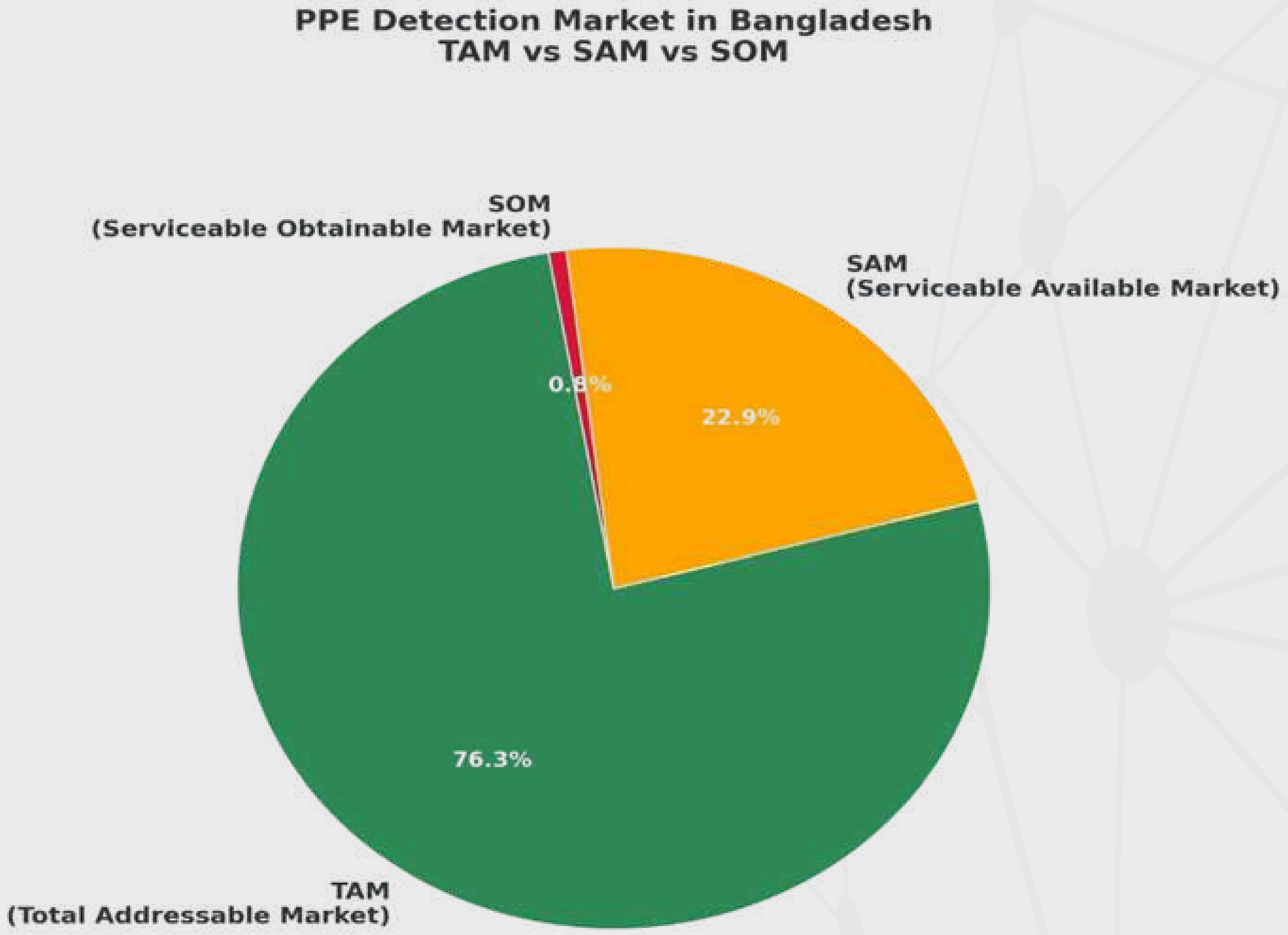


Figure 1 PPE Detection Market Segmentation in Bangladesh: TAM vs SAM vs SOM

Industry Sector	Typical Safety-Critical Areas	PPE Requirements
Pharmaceutical	Clean rooms, chemical labs	Masks, gloves, goggles, gowns
Food & Beverage	Processing & packaging lines	Gloves, masks, aprons
Construction	Building sites, scaffolding areas	Helmets, vests, boots, gloves
Textile & Garments	Dyeing units, chemical treatment areas	Gloves, masks, safety glasses
Manufacturing (General)	Assembly lines, machinery operation areas	Hard hats, gloves, ear protection
Logistics & Warehousing	Loading docks, cold storage, transport hubs	Safety vests, steel-toe boots, gloves, helmets

Table 1 PPE Requirments

Global PPE Market Overview:

- Rapid growth driven by workplace safety, compliance, and computer vision.
- USD **311M in 2023,** projected to reach USD **117.4B by 2033** (CAGR: 81%) – Market.us report.
- USD 1**.8B in 2024**, growing to USD **358.6B by 2033** – Straits Research report.
- Reflects the massive potential for AI-driven safety systems.

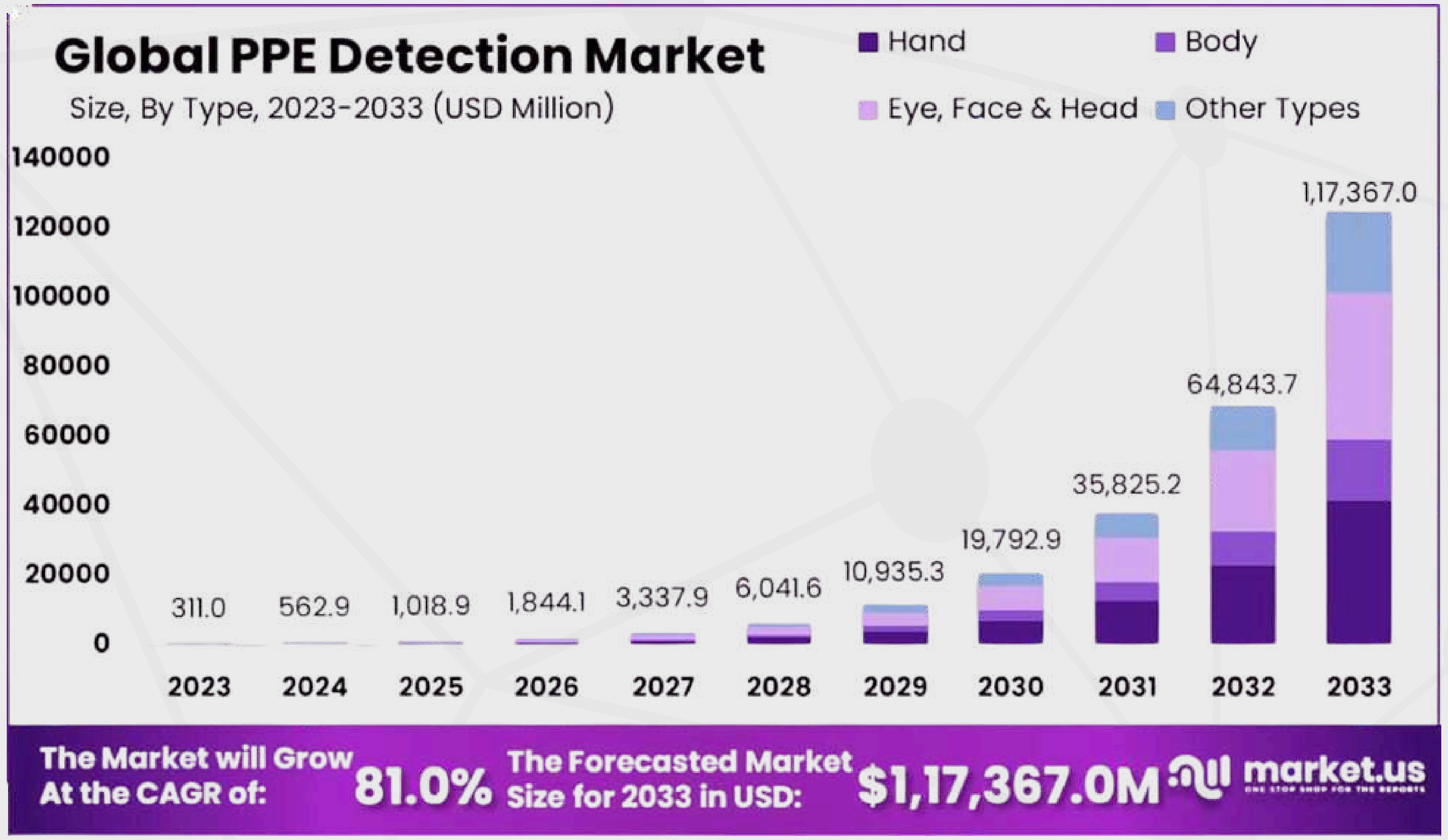


Figure 2 Projected Global PPE Detection Market Size by Segment, 2023–2033 (USD Million)

Bangladesh Market Insights

PPE demand is rising due to industrial growth, with smart and disposable healthcare PPE expected to grow significantly from 2025 to 2031.

Key Metric	Value / Insight
Number of Construction Workers	3.5 million+ (as of 2023)
Annual Growth Rate	6.5% (steady growth driven by urbanization and infrastructure projects)
Workplace Accident Rate	Highest among all industries (ILO & DIFE reports)
Estimated PPE Detection Market (Construction)	USD 1.2 – 1.5 million (SOM estimate)
Major Project Hubs	Dhaka, Chattogram, Rajshahi, Khulna, Sylhet
Common PPE Needed	Helmets, vests, gloves, boots

Table 2 Industry-Specific Safety-Critical Areas and Corresponding PPE Requirements

Similar Products

Company	Product Name	Price (Taka)
Visionify	VisionPlatform PPE Detection	85,000 (one-time)
Google	Vertex AI PPE Model	5,000/month(Subscription)
Protex AI	Protex PPE Monitoring Platform	1,20,000/year(Subscription)
Intenseye	Intenseye PPE Detection Suite	1,50,000/year(Subscription)

Table 3 Comparison of Similar PPE Detection Products and Their Pricing

SWOT Analysis:

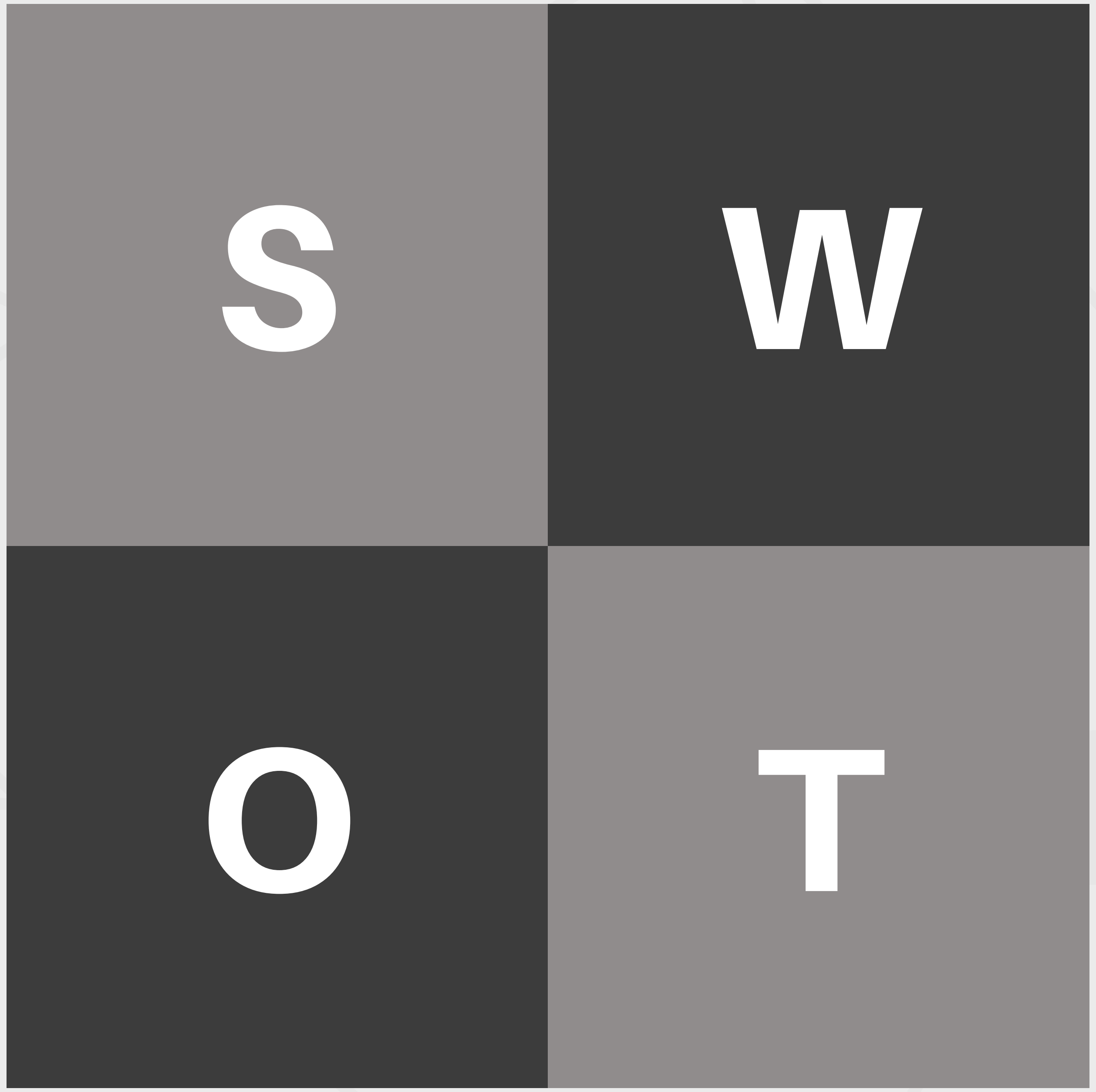
Product	Strengths	Weaknesses	Opportunities	Threats
Vertex AI PPE Detection	1. Backed by DeepMind with cutting-edge AI research and resource 2. Built on Google Cloud's scalable, enterprise-grade Vertex AI platform 3. Likely integrates well with existing clouds infrastructure and workflows.	1.Not tailored specifically for PPE -more of a general ML platform that requires significant customization. 2.Might lack real-time edge inference, leading to latency. 3.High costs.	1. Large Enterprise users can unify PPE detection with broader analytics pipelines. 2. Opportunity to add value via pertained models, AutoML and MLops	1.Highly technical—requires in-house ML talent or consulting support. 2.Specialized competitors (e.g. Intenseye) offer ready-to-use PPE detection.
Protex	1.Self-serve and highly customizable-suits small to mid-market clients. 2.Modular approach allows tailoring to specific PPE items or workflows.	1.Complexity and heavy customization may hamper scalability in large enterprises. 2.Not industry-specific—potentially inconsistent accuracy across environments.	1. Could expand support with industry templates or marketplace models. 2. Offer vertical bundles e.g. for manufacturing.	1.Specialized turnkey systems outcompeting on ease-of-use and reliability. 2.Customization overhead may deter time-sensitive clients.
Intenseye	1.Purpose-built PPE compliance module, covering ~10 PPE types with high precision. 2.Quick deployment over existing CCTV (cloud, hybrid, on-prem). 3.. Integrates into EHS workflows with dashboards, scoring, automated corrective alerts.	1.Likely subscription cost may be prohibitive for smaller operations. 2.Dependence on camera placement and visual conditions. 3.High Price	1. Expand into new industries (construction, logistics). 2. Add fine-grained detection (correct fit, sequence).	1.Edge-processing challengers or localized inference competitors. 2.New solutions using vision-language models may offer richer context.
VisionPlatform	1.Offers free standard PPE models and full control via re-training. 2. Highly flexible: supports various camera types (RGB, thermal, lidar). 3. Edge/on-premise deployment ensures low latency and privacy.	1.Requires technical know-how—users must annotate data and retrain models. 2.Baseline model accuracy may be inconsistent in real-world conditions.	1.Can build a community for reusable PPE modules or datasets. 2.Offer managed services or consult to reduce deployment friction.	1.Easy-to-use commercial plug-and-play systems (e.g. Intenseye) may capture non-technical users. 2.Risk of low adoption if retraining burden is too high.
Our Product	1.Low price 2.Quick installation (Plug and play) 3.Can be used with existing service	1.New player, less trust at first, limited funding 2.Less-powerful hardware. 3.No cloud service support	1.First mover advantage at local market 2.Price to performance is attractive to customer	1.Intense competition of global competitor 2.Regulatory challenges meeting diverse international standard

Table 3 Comparison of Similar PPE Detection Products and Their Pricing

SWOT ANALYSIS

STRENGTHS

- Low price
- Quick installation (Plug and play)
- Can be used with existing service



WEAKNESSES

- New player, less trust at first, limited funding
- Less powerful hardware.
- No cloud service support

OPPORTUNITIES

- First mover advantage at local market
- Price to performance is attractive to customer

THREATS

- Intense competition of global competitor
- Regulatory challenges meeting diverse international standard

Market Survey

Customer Requirements Analysis Report for AI-Based PPE Detection System

Objective:

- **1.** Understand the market needs for AI-based PPE Detection Systems in Bangladesh's industrial sector.
- **2.** Gather both numerical (quantitative) and opinion-based (qualitative) data.
- **3.** Use this data to guide product development—such as features, technical needs, and pricing.

Van Westendrop Price Sensitivity meter:

1) What price would you consider too expensive to pay for this PPE-detection camera?

2) What price would you consider too low to doubt good quality for this system?

3) What price is high but still acceptable for you?

4) At what price would you consider cheap but the quality would be maintained?

The Van Westendorp Price Sensitivity Meter analysis identifies 6800 BDT as the optimal price for the AI-based PPE Detection System, where perceptions of "too cheap" and "too expensive" intersect. The acceptable price range is 3950–8750 BDT, with 6900 BDT as the indifference point. Pricing between 6800–6900 BDT offers the best balance of value and acceptance, positioning the product as reliable, high-quality, and suitable for cost-sensitive industries.

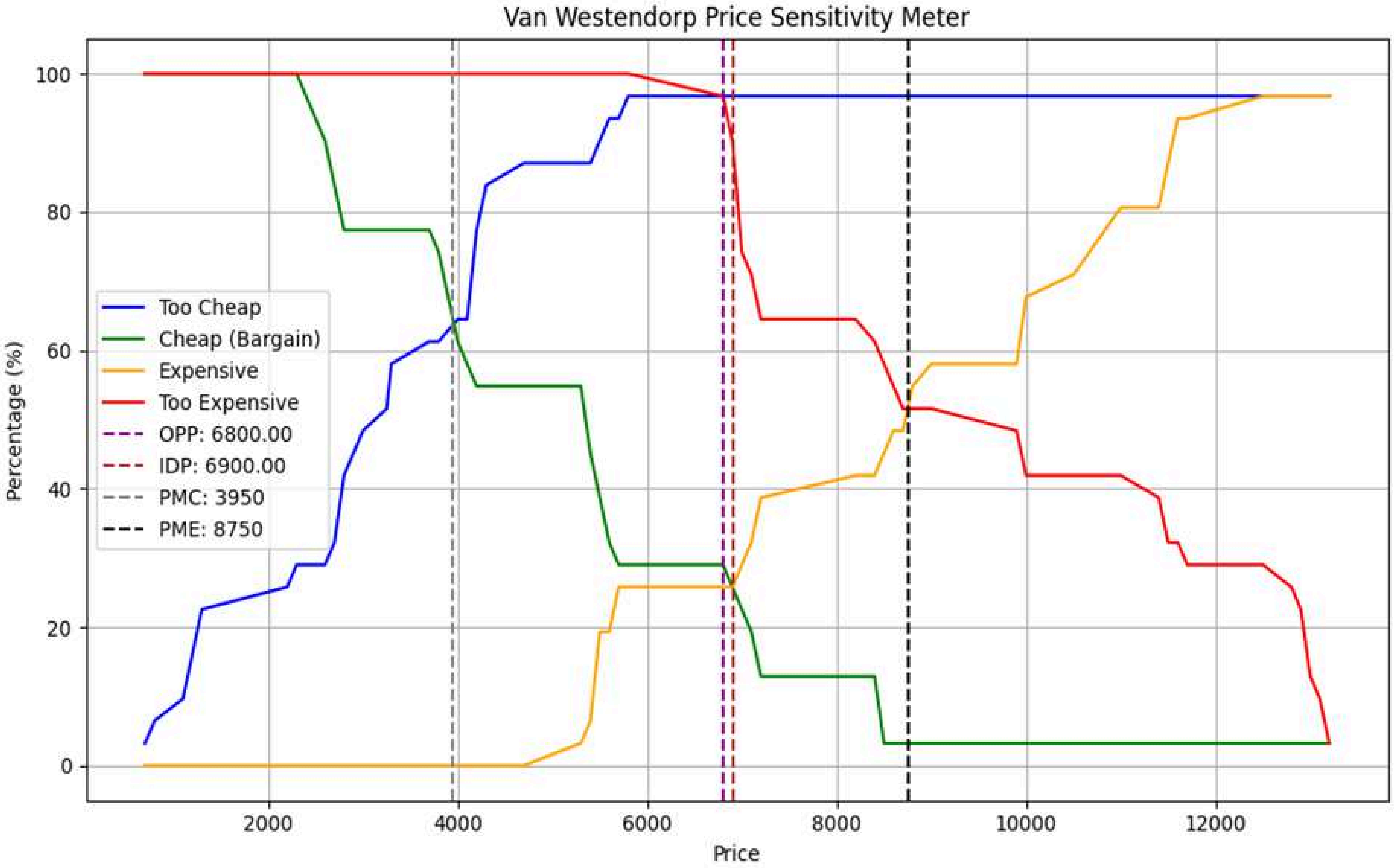


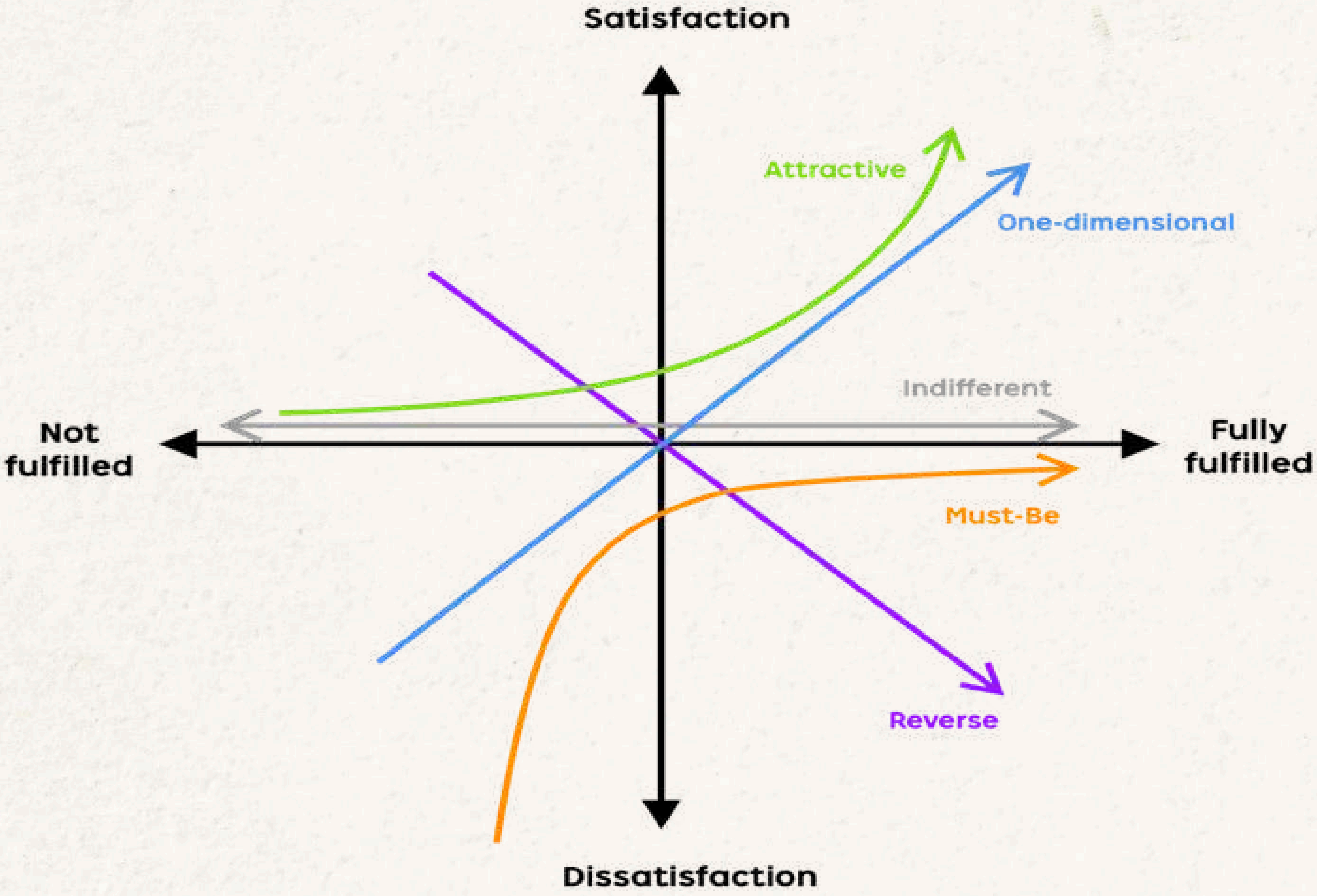
Figure : Van westendrop Price Sensitivity Meter

Kano Model Need Assesment

The Kano Model helps prioritize product features based on their impact on customer satisfaction. It classifies features into:

1. Basic Needs – Expected essentials; missing them causes dissatisfaction.
2. Performance Needs – More is better; directly increase satisfaction.
3. Excitement Needs – Unexpected delights; boost satisfaction if present, but not missed if absent.
4. Indifferent – Features that don’t affect satisfaction.
5. Reverse – Features liked by some but disliked by others.

KANO Model Diagram



Kano Evaluation Matrix

Service quality elements		Dysfunctional				
		Like	Expect	Neutral	Accept	Dislike
Functional	Like	Q	A	A	A	O
	Expect	R	I	I	I	M
	Neutral	R	I	I	I	M
	Accept	R	I	I	I	M
	Dislike	R	R	R	R	Q

(Note: A: Attractive; O: one-dimensional; M: must-be; I: indifferent; R: reverse; Q: questionable.)

Table 1: Example Kano Model Need Assessment Questions

Customer	With Real time alerts (Functional)	Without Real time alerts (Dysfunctional)	Feature
Customer1	Must be	Dislike	Basic
Customer2	Must be	Like	Reverse
Customer3	Like	Dislike	One dimensional
Customer4	Must be	Dislike	Basic
Customer5	Must be	Live with	Basic

Here is a list of the customer requirements:

1. Real Time Alert
2. Accuracy
3. Multiple Detection
4. Maintenance Cost
5. Distance
6. Multiple Language
7. Long Time Use
8. Colour
9. Mobile Notifications
10. Battery or Secondary Power Option
11. Integration with Existing Safety Monitoring System
12. Automatic Report Generation
13. Quick Response When Someone Is Not Wearing PPE
14. Works in Both Indoor and Outdoor Construction Environments
15. Functionality in Low Internet Zones
16. Portability of the Product

Customer Requirements	Must be	One Dimensional	Attractive	Indifferent	Reverse	Uncategorized	Max Category
Real Time Alert	70%	20%	5%	5%			Must Be
Accuracy	63.64	22.73%	9.09%			4.55%	Must Be
Multiple Detection	25	15.00%	55.00%	5.00%			Attractive
Maintanance Cost	10%	50%	30%		5%		One Dimensional
Distance	30%		30%	15%	5%	5%	Must be
Multiple Language	10%	15%	65%	5%	5%		Attractive
Long Time Use	50%	18.18%	9.09%			22.73%	Must be
Colour		9.09%	9.09%	77.27%		4.55%	Indifferent
Mobile notifications	13.64%	13.64%		18.18%	27.27%	13.64%	Reverse
Battery Or Secondary Power Option	4.55%	9.09%	27.27%	13.64%		45.45%	Attractive
Integration with existing safety monitoring system	20.00%	20.00%	45.00%	10.00%	5.00%		Attractive
Automatic report generation	22.73%	18.18%	22.73%	31.82%		4.55%	Indifferent
Quick Response When Someone Is Not Wearing PPE	50.00%	25.00%	20.00%	5.00%			Must be
Works in both indoor and outdoor construction environments	47.62%	28.57%	4.76%	19.05%			Must be
Functionality in low internet zones	40.91%	40.91%	9.09%	9.09%			Must be
Portability Of The Product	22.73%	9.09%	4.55%	9.09%	4.55%	50.00%	Must be

Reliabilty

For reliability, we used the statistical tool Cronbach’s Alpha to measure the internal consistency.

Cronbach’s alpha is a statistical measure used to evaluate the internal consistency or reliability of a set of survey or test items.

Number of items

$$\alpha = \frac{N\bar{c}}{\bar{v} + (N-1)\bar{c}}$$

Average variance

Average inter-item covariance among the items

Formula of Cronbach’s Alpha

$$\alpha = 0.617$$

Cronbach's Alpha Score	Level of Reliability
0.0 – 0.20	Less Reliable
>0.20 – 0.40	Rather Reliable
>0.40 – 0.60	Quite Reliable
>0.60 – 0.80	Reliable
>0.80 – 1.00	Very Reliable

Reliabilty measure of Cronbach's alpha

QUALITY FUNCTION DEVELOPMENT(QFD)

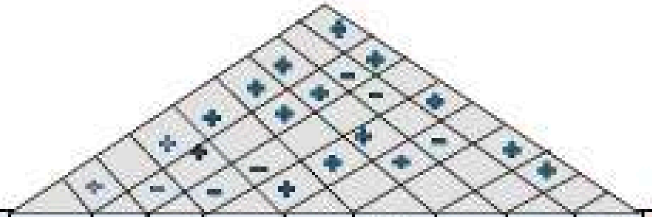
Quality Function Deployment (QFD) is a method used to transform customer requirements (often called the "voice of the customer") into engineering characteristics for a product. The **"House of Quality"** is the most recognized form of QFD, and it helps to prioritize and align technical specifications with customer needs. It is to be noted that the relative weight is determined on the scale of 1 to 9

Table 3: Symbols to illustrate correlations between technical requirements

Types of correlation	Symbol
Strongly Positive	++
Positive	+
No Correlation	Blank
Negative	-
Strongly Negative	--

Table 4: Numbers to define relationship between the customer and technical requirements

Types of correlation	Symbol
No Relation	Blank
Weak Relation	1
Moderate Relations	3
Strong Relation	9



Serial No	Customer priority	Functional Requirements (How) → Customer Requirements (What) ↓	AI Model with software	Camera	Battery	Processor	Alert System	Detection Frame Rate	Accuracy Rate	Weight	Detection Range	Competitor Comparison: Goggle Vertex AI	Competitor Comparison: Hikvision	Competitor Comparison: Visionify
1	10	Real Time Detection	9	9		9		9	9		9	3	5	4
2	10	Accuracy	9	9		9		9	9		3	4	4	5
3	9	Multiple Detection	9	9		9		9	3		3	4	3	4
4	8	Maintenance Cost	3	3	3	9	3			9		2	4	3
5	8	Distance	9	9		9		3	3		3	5	4	4
6	3	Multiple Language	3						5			4	4	5
7	9	**Battery / Backup Power**	8	3	9	9	3	1				5	2	3
8	8	Long Time Use	9	9	3			3		3		5	3	4
9	8	System Integration	3									4	3	4
10	8	PPE Detection Alert	9	9		9	9	9			9	3	5	4
11	9	Indoor & Outdoor Use	9	9				9	5		3	3	4	4
12	8	Low Internet Functionality	9		9			9				2	4	4
13	8	Portable Design		9	9					9		5	4	4
		Technical Importance Rating	537	681	291	558	195	543	231	168	270			
		Goggle Vertex AI	Vertex AI Visio			Intel Xeon	Email/SMS	1	85%-95%	8.55 Kg	15-20m			
		Hikvision	GuanlanAIoT	8 MP	20 ah	Hisilicon SoCs	Speakers	25	90%-98%	_10 Kg	8-15m			
		Visionify	YOLO	12 mp	4p ah	Intel Core i7	Speakers	20	95%-99%	10 Kg	5-12m			
		Target	YOLO	10 mp	10000 mah	Rasbery pi	LED bulb beep sound	30	90-99%	5 Kg	3-4m			
		Priority Rank	4	1	5	2	8	3	7	9	6			

Figure 27: House of Quality for AI Based PPE Detection System

Design and Specification Analysis

Introduction: Design and Specification Analysis is a key phase in product development, where a proposed design is evaluated for functionality, feasibility, performance, and cost. It ensures the design meets customer needs, technical limits, and regulatory standards, while optimizing quality and minimizing risks.

Specification Chart:

Table 01: Electrical Components

Parts Name	Description	Quantity
Raspberry Pi 3B+	Main processor	1
Raspberry Pi Camera Module v2	Image input	1
MicroSD Card (16–32 GB)	Storage for OS & model	1
5V Buzzer (Active)	Beep alert	1
Red LED (5mm)	PPE violation alert	1
Green LED (5mm)	PPE OK alert	1
220Ω Resistors	For LED current limiting	2
GPIO Header Wires	For LED/Buzzer connection	~6
Power Bank (10,000 mAh)	Secondary power	1
Micro-USB Cable	Power connection	1
Mini Breadboard / PCB	For mounting components	1

Table 02: Structural Components

Parts Name	Description	Quantity
Case or Mount (Custom)	3D printed/assembled housing	1

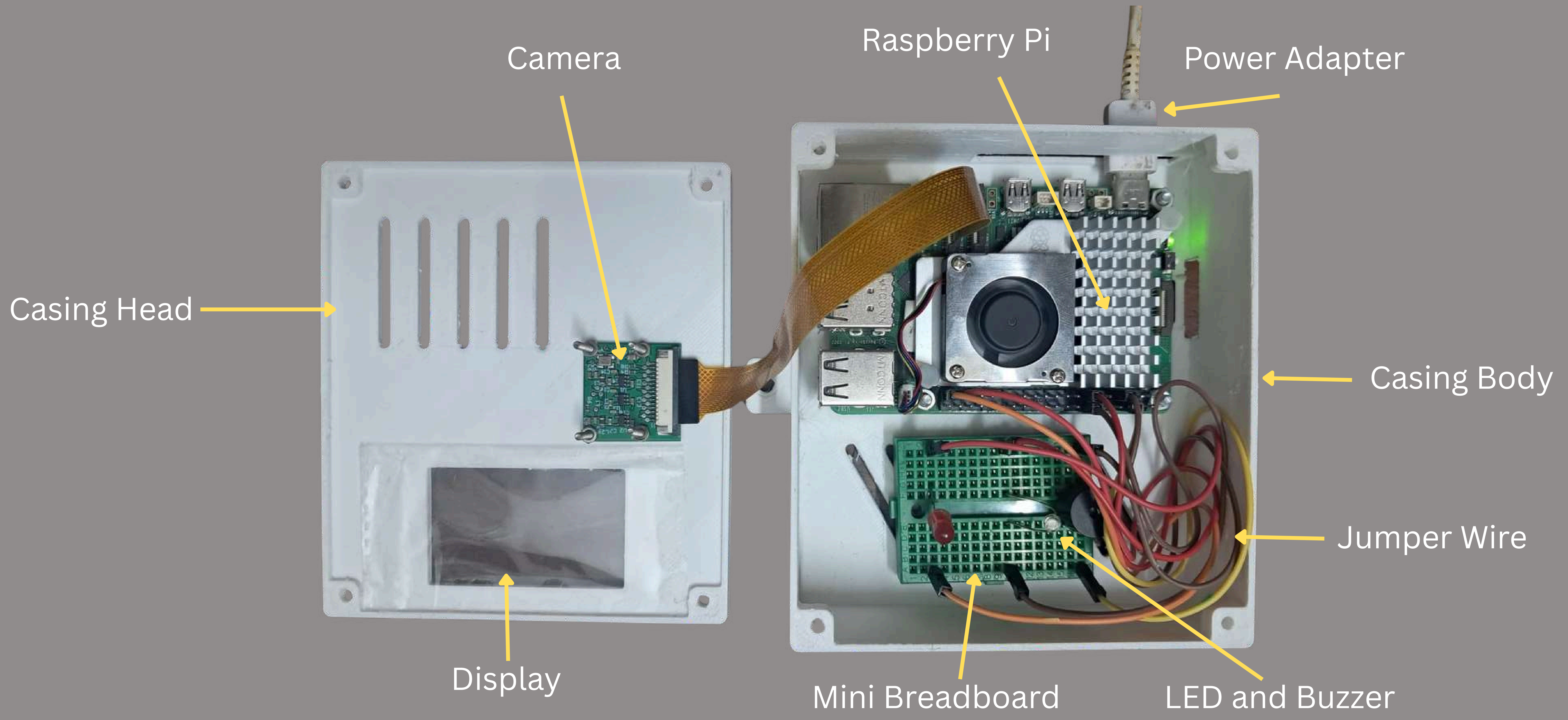
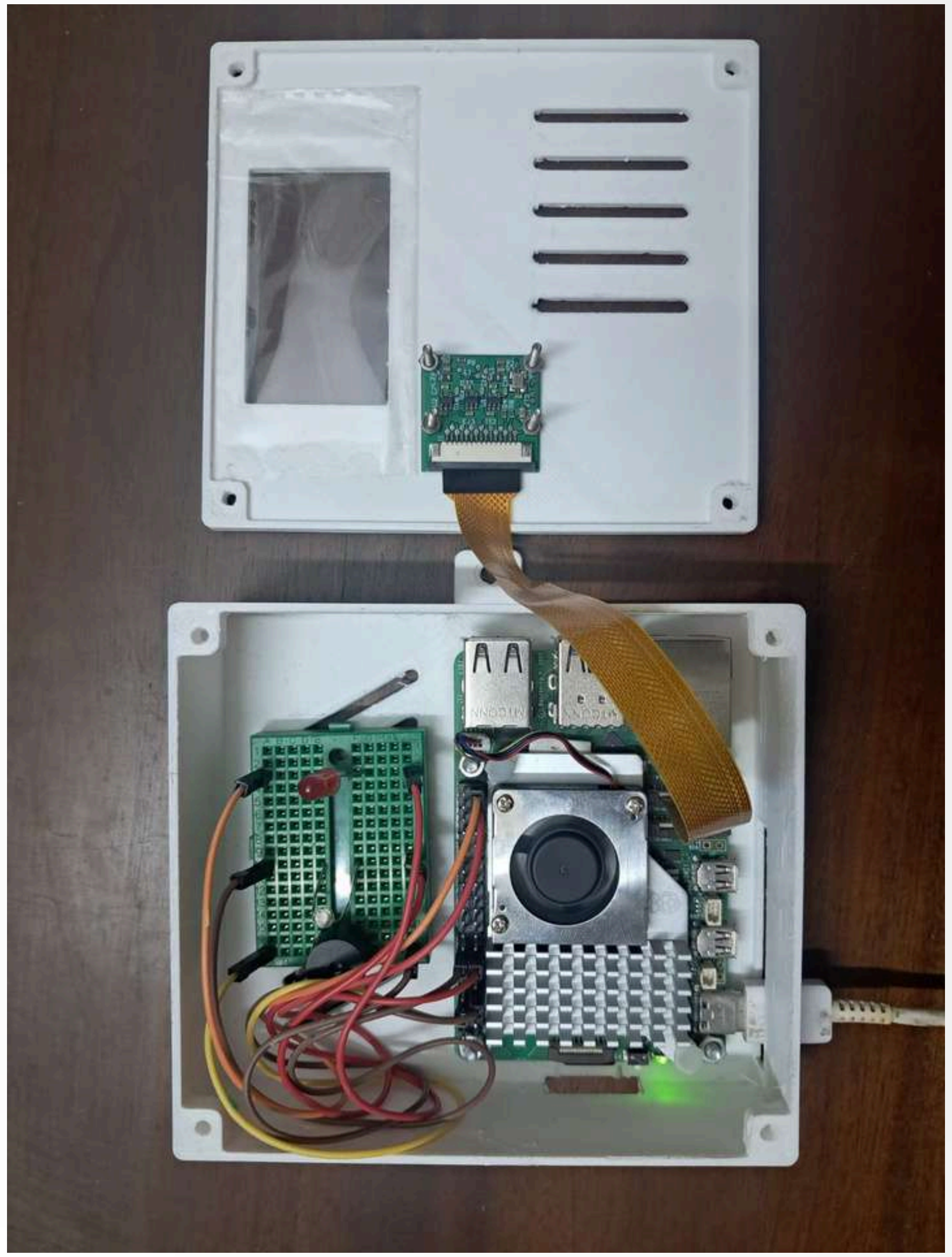


Figure: Structural and Electrical Components of the product

PROTOTYPE

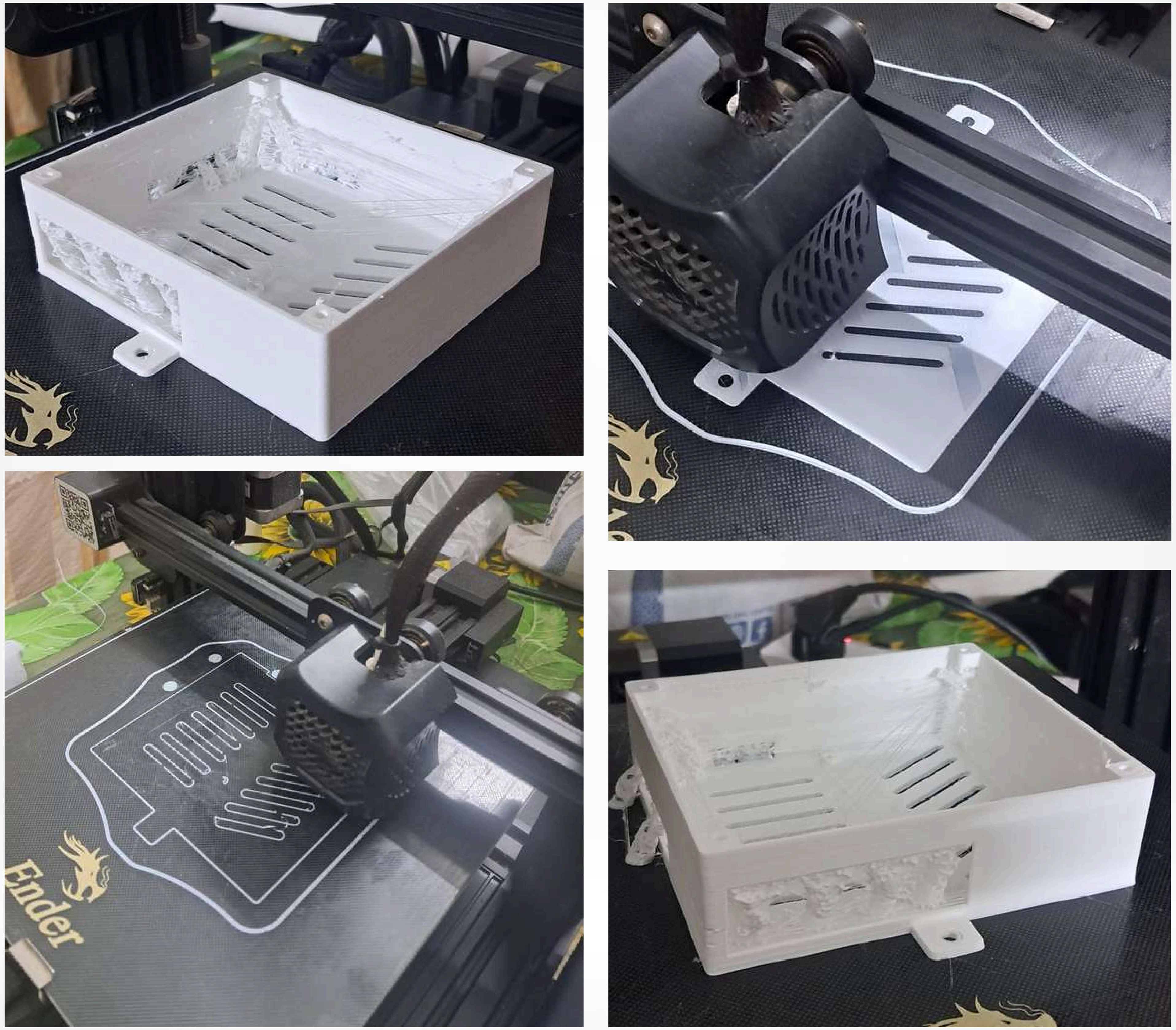
PROTOTYPE

ASSEMBLED VIEW



INSIDE VIEW

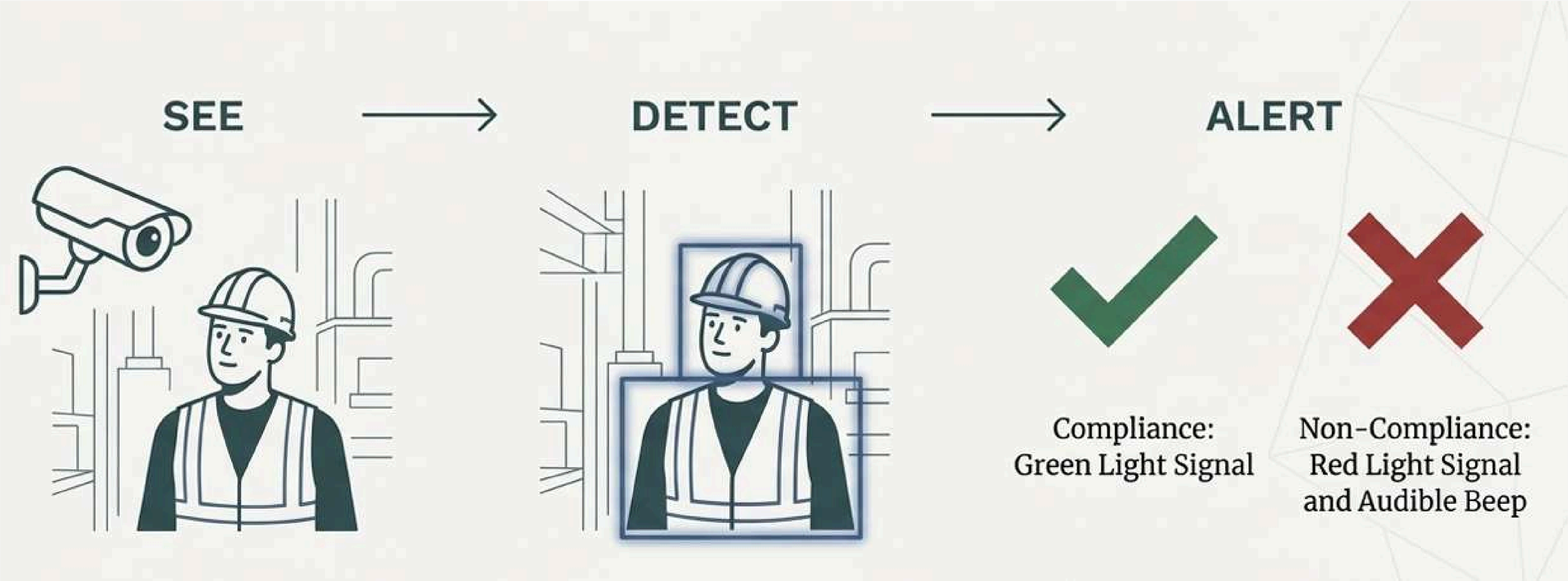
PROTOTYPE



MANUFACTURING PROCESS OF THE PROTOTYPE

PROTOTYPE WORKING MECHANISM

THE SYSTEM USES A PROVEN MACHINE LEARNING ALGORITHM (YOLO – YOU ONLY LOOK ONCE) TO ANALYSE VIDEO FEEDS IN REAL TIME.



CONCLUSION

- Enables real-time, automated PPE compliance monitoring to improve workplace safety
- Market-driven design validated through surveys, Kano Model, and QFD
- Provides high accuracy, instant alerts, and offline operation
- Features a simple, reliable functional structure (sense–process–alert)
- Demonstrates strong technical feasibility and industrial applicability

Thank You